Telecom Customer Churn Analysis

End-to-End Analytics Project

Stage 1: Excel — Data Preprocessing & Exploratory Analysis

1. Project Overview

This project performs a full end-to-end customer churn analysis for a telecom company using three tools: Excel for preprocessing and exploratory analysis, SQL for deep querying and insight extraction, and Power BI for interactive dashboard and visualization.

Objective
Identify the key drivers of customer churn, quantify revenue impact, and deliver actionable business recommendations to reduce churn and improve customer retention.

2. Dataset Description

Source: Telecom Customer Churn Dataset (California, Q2 2022)

Attribute	Detail
Total Records	7,043 customers
Total Columns	38 columns
Period	Q2 2022
Geography	California, USA
Target Variable	Customer Status (Churned / Stayed / Joined)
Churn Rate	26.5% (1,869 churned out of 7,043)

3. Stage 1 — Excel: Data Preprocessing & EDA

Excel was used as the first layer of the analysis pipeline — responsible for data cleaning, column engineering, and preliminary exploratory analysis via pivot tables.

Step 1: Handling Missing Values

A thorough check of all columns was performed using Excel's Filter function to identify blank cells. The following columns contained blanks and were treated accordingly:

Group 1 — No Phone Service
Customers without phone service had blanks in phone-related columns.

- Avg Monthly Long Distance Charges — replaced blanks with 0 (numeric, no charge applicable)
- Multiple Lines — replaced blanks with 'No Phone Service'

Group 2 — No Internet Service
1,526 customers had no internet service, causing blanks across all internet-related columns.

- Internet Type — replaced with 'No Internet Service'
- Avg Monthly GB Download — replaced with 0
- Online Security, Online Backup, Device Protection Plan — replaced with 'No Internet Service'
- Premium Tech Support, Streaming TV, Streaming Music, Unlimited Data — replaced with 'No Internet Service'

Group 3 — Not Applicable
Churn-related columns were blank for customers who did not churn — this is expected behaviour.

- Churn Category — replaced blanks with 'Not Applicable'
- Churn Reason — replaced blanks with 'Not Applicable'

Note: The Offer column contained 'None' as a text value (not blank) — no replacement was needed.

Note: The dataset was originally protected. A clean copy was created by copying all data and pasting as values only into a new sheet — preserving the raw data and following best practice.

Step 2: Age Group Column (Calculated Column)

A new column 'Age Group' was created next to the Age column to segment customers into meaningful age brackets for analysis.

Formula Used
=IF(C2<=30,"18-30",IF(C2<=45,"31-45",IF(C2<=60,"46-60","61+")))

This nested IF formula evaluates the Age value and assigns one of four groups:

- 18–30 — Young adults
- 31–45 — Middle-aged adults
- 46–60 — Senior adults
- 61+ — Elderly customers

Step 3: Tenure Group & Customer Category Columns (Calculated Columns)

Two new columns were created to segment customers by their length of relationship with the company:

Tenure Group Formula
=IF(F2<=12,"0-1 Year",IF(F2<=24,"1-2 Years",IF(F2<=48,"2-4 Years","4-6 Years")))

Customer Category Formula
=IF(F2<=12,"New Customer",IF(F2<=24,"Developing Customer",IF(F2<=48,"Established Customer","Loyal Customer")))

This produces four customer lifecycle categories:

Tenure Group	Customer Category	Rationale

0–1 Year	New Customer	Highest churn risk — still evaluating service
1–2 Years	Developing Customer	Building loyalty — still vulnerable
2–4 Years	Established Customer	More stable — moderate churn risk
4–6 Years	Loyal Customer	Lowest churn risk — strong relationship

Step 4: Pivot Table EDA

Three pivot tables were built in a dedicated 'Pivot Analysis' sheet to extract preliminary insights from the cleaned data.

Pivot Table 1 — Churn by Contract Type

Contract Type	Churned	Joined	Stayed	Grand Total	Churn Rate
Month-to-Month	1,655	408	1,547	3,610	45.8%
One Year	166	24	1,360	1,550	10.7%
Two Year	48	22	1,813	1,883	2.5%
Grand Total	1,869	454	4,720	7,043	26.5%

Insight: Contract type is the strongest churn predictor

Month-to-Month customers churn at 45.8% — nearly 18x higher than Two Year contract customers (2.5%). The longer the commitment, the stronger the retention.

Recommendation: The company should aggressively incentivize Month-to-Month customers to upgrade to longer contracts through targeted discounts, loyalty rewards, or bundled offers.

Pivot Table 2 — Churn by Internet Type

Internet Type	Churned	Joined	Stayed	Grand Total	Churn Rate
Fiber Optic	1,236	101	1,698	3,035	40.7%
Cable	213	56	561	830	25.7%
DSL	307	115	1,230	1,652	18.6%
No Internet Service	113	182	1,231	1,526	7.4%
Grand Total	1,869	454	4,720	7,043	26.5%

Insight: Fiber Optic customers churn at an alarming rate

40.7% of Fiber Optic customers churned — despite it being the premium internet service. This suggests the pricing or service quality is not meeting customer expectations relative to competitors.

Recommendation: Urgently review Fiber Optic pricing and service quality. Deploy targeted retention campaigns for Fiber Optic customers on Month-to-Month contracts — these represent the highest combined risk segment.

Pivot Table 3 — Average Monthly Charge by Customer Status

Customer Status	Avg Monthly Charge
Churned	$73.35
Stayed	$61.74
Joined	$42.78
Overall Average	$63.60

Insight: Churned customers paid the highest monthly charges

Customers who churned were paying $73.35/month on average — $11.61 (18.8%) more than customers who stayed. This confirms that high-paying customers feel the least value for money and are most likely to switch to competitors.

Recommendation: Introduce a loyalty pricing model or value-add perks for high-paying customers. A $10/month discount for long-tenure Fiber Optic customers could significantly improve retention ROI.

4. Stage 1 Summary

What was achieved in Excel

Data cleaned and standardized across 7,043 records | Missing values handled logically across 13 columns | 3 calculated columns created (Age Group, Tenure Group, Customer Category) | 3 pivot tables revealing key churn patterns | 3 business insights identified and documented

Next Stage: SQL — Deep querying, churn rate calculations, revenue impact analysis, and high-risk customer profiling.